10/10

MATHEMATICS 2210

Quiz ~~2~~ 4, Fall semester 2012

Name: KEY

3 pts

1. Compute
$$\frac{\partial^2}{\partial x \partial y}(\ln(x^2-y^2)).$$

$$\frac{\partial}{\partial y} \ln(x^2-y^2) = \frac{-2y}{x^2-y^2}$$

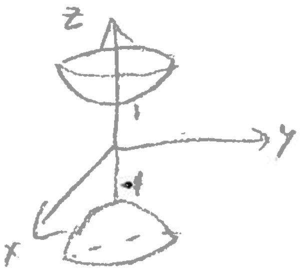
$$\frac{\partial}{\partial x}\left(\frac{-2y}{x^2-y^2}\right) = -2y \cdot \frac{-2x}{(x^2-y^2)^2} = \frac{4xy}{(x^2-y^2)^2}$$

4 pts

2. Sketch the level surfaces of $w = x^2 + y^2 - z^2$ for $w = -1, 0, 1$.

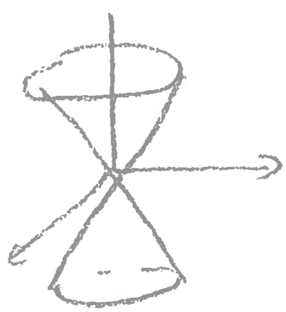
$w = -1 = x^2+y^2-z^2$

$x^2+y^2 = z^2-1$



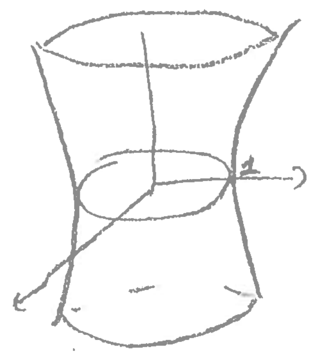
$w = 0 = x^2+y^2-z^2$

$x^2+y^2 = z^2$

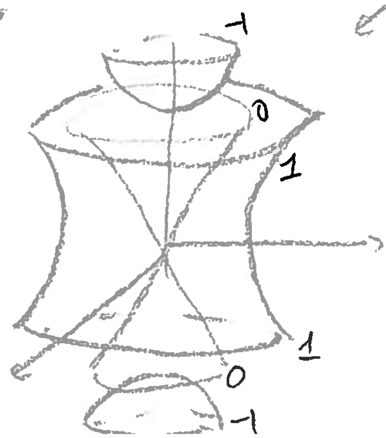


$w = 1 = x^2+y^2-z^2$

$x^2+y^2 = z^2$



or



3 pts

3. Change $x^2 + y^2 - z^2 = 1$ to cylindrical and spherical coordinates.

cylindrical $x^2+y^2=r^2$ $z=z$ $r^2-z^2=1$

spherical $x^2+y^2+z^2=\rho^2$ $x^2+y^2=\rho\sin\phi$

$1=x^2+y^2-z^2=2(x^2+y^2)-(x^2+y^2+z^2)=2\rho^2\sin^2\phi-\rho^2$

$=\rho^2(2\sin^2\phi-1)$

$1=\rho^2(\sin^2\phi-\cos^2\phi)$ $\cancel{=\rho^2\cos^2\phi}$

Do not need to simplify

$1=\rho^2\sin^2\phi\cos^2\theta+\rho^2\sin^2\phi\sin^2\theta-\rho^2\cos^2\phi$

is OK